

ANKIT ADDYA

Data Analyst | Business Analyst | SQL, Python, Power BI, Tableau, Snowflake | AI-Enabled Analytics |
FinTech and Financial Services

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PROFESSIONAL SUMMARY

Data Analyst and Business Analyst with 3 years of professional experience in FinTech SaaS and Business Consulting, specialising in AI-enabled analytics, executive dashboard development, and translating ambiguous business problems into structured analytical frameworks that drive commercial decisions. Uniquely combining core DA and BA skills in SQL, Python, Power BI, Tableau, Snowflake and stakeholder communication with production-grade AI tooling that automates insight delivery, enables natural language data investigation, and brings predictive intelligence directly into analytical workflows. Track record of scoping business requirements with cross-functional stakeholders, defining KPI frameworks, improving fraud detection accuracy, reducing reporting turnaround, and delivering clear, actionable recommendations to senior and VP-level leadership through rigorous hypothesis testing, A/B experimentation, and root cause analysis.

TECHNICAL SKILLS

SQL and Data Platforms: Advanced SQL (CTEs, Window Functions, Complex Joins, Query Optimisation, Partitioning, Composite Indexes), Snowflake, PostgreSQL, BigQuery, Databricks SQL, MySQL, dbt, PySpark

Python: Pandas, NumPy, SciPy, scikit-learn, Matplotlib, Seaborn, Plotly, Data Wrangling, EDA, Automation Scripting, ETL Pipeline Development

Power BI and Tableau: Power BI (DAX, DirectQuery, Row Level Security, Time-Intelligence: YTD, QTD, MoM, Rolling 3M and 12M, Conditional Formatting, Drill-Through, Parameters, Self-Serve Reporting), Tableau (LOD Expressions, Calculated Fields, Parameter Controls, Dashboard Actions), Amazon QuickSight, Streamlit

Business Analysis and Analytics: Business Requirements Gathering, Use Case and User Story Definition, Stakeholder Scoping and Prioritisation, Gap Analysis, KPI Framework Design and Monitoring, Root Cause Analysis, A/B Testing and Hypothesis Testing (ANOVA, Chi-Squared, t-Test, Correlation), Funnel Analysis, Cohort Analysis, Conversion and Retention Analytics, Revenue and Growth Analytics, Opportunity Sizing, Predictive Modelling (XGBoost, LightGBM), Customer Segmentation, Customer Lifetime Value Modelling, Churn Prediction, NLP and Text Classification, Anomaly Detection, SHAP

AI-Enabled Analytics and Automation: Natural Language Data Investigation (LangGraph, LangChain), AI-Assisted Insight Generation (Azure OpenAI GPT-4o, Gemini, Vertex AI), RAG Pipelines for Self-Serve Knowledge Retrieval, Prompt Engineering, Model Context Protocol (MCP), Power Automate, n8n Workflow Automation, Responsible AI (Human Validation, Bias Detection, PII Sanitisation)

Data Engineering and Collaboration: ETL and ELT Pipeline Design, Apache Airflow, Databricks, Delta Lake, Medallion Architecture, Amazon S3, Azure ADLS Gen2, Great Expectations, Schema Validation, Data Dictionary and Lineage Documentation, Advanced Excel (VBA, Pivot Tables, Power Query, Array Formulas, INDEX-MATCH), Agile, Jira, Cross-Functional Collaboration, Technical Documentation, Git

WORK EXPERIENCE

Independent Data Science and Analytics Practitioner | Self-Directed

Nov 2022 to Present | Kolkata, India

Structured self-directed upskilling period combining Scaler DSML Program (Feb 2023 to Jun 2025) with independent development of 5 production analytics platforms, BI dashboards, data pipelines, and AI-enabled insight tools across BFSI, retail, and commercial analytics domains.

- **Scoped business problems into structured analytical briefs with defined requirements and success criteria:** translated complex commercial questions across customer value, fraud risk, churn, and revenue optimisation into concrete analytical workstreams — each with documented data requirements, methodology assumptions, metric definitions, and stakeholder-facing deliverable specifications before a single line of code was written.
- **Built AI-enabled analytics workflows that accelerate insight delivery for business stakeholders:** deployed production LangGraph AI agents on GCP Vertex AI enabling natural language data investigation by non-technical stakeholders,

automated insight narrative generation via Azure OpenAI grounded in deterministic model outputs, and RAG pipelines achieving over 95 percent contextual recall for self-serve knowledge retrieval. Participated in the HuggingFace MCP Hackathon.

- **Defined KPI frameworks and built executive dashboards for commercial decision-making:** Power BI dashboards with 30 plus DAX measures, time-intelligence, traffic-light conditional formatting, and self-serve filtering for leadership reporting. Tableau visualisations with LOD expressions and parameter controls. Streamlit self-serve applications enabling non-technical stakeholders to independently interrogate data.
- **Completed a 28-month structured DSML program at Scaler** with module-gated written examinations and technical interviews at each stage. Mentored by practitioners from Microsoft, Walmart, Morgan Stanley, and Ola Electric.

Research Analyst | Data Analytics and Business Intelligence | Decimal Point Analytics Jun 2022 to Nov 2022 | Nashik, India

Business Consulting and FinTech SaaS | FidelityPulse (fraud detection) and CreditPulse (NLP credit risk classifier)

- **Scoped and defined analytical requirements with FinTech clients and internal stakeholders:** led discovery conversations with client business owners to translate ambiguous fraud and credit risk challenges into structured analytical briefs, defining scope, data requirements, methodology assumptions, acceptance criteria, and success metrics before project execution — the core BA deliverable that prevented misaligned analytical effort on both FidelityPulse and CreditPulse.
- **Improved fraud detection model accuracy from 85 percent to 92 percent** by engineering 21 behavioural and transactional features on Databricks PySpark, applying hypothesis-driven feature selection using ANOVA and chi-squared analysis, and validating through rigorous cross-validation, directly reducing compliance operational costs for the FinTech SaaS client.
- **Reduced SQL analysis runtime by 40 percent** through advanced query optimisation (CTEs, window functions, composite indexes) across large transactional databases in Snowflake and Databricks, accelerating insight delivery cycles to client leadership.
- **Built Power BI and Tableau executive dashboards** providing real-time KPI visibility to client leadership across product, compliance, and commercial functions, translating complex analytical findings into clear business narratives and actionable recommendations.
- **Spearheaded the automation of data cleaning pipelines using Power Automate**, seamlessly routing operational inputs to the data warehouse and reducing data preparation time from days to minutes.
- **Built the CreditPulse NLP credit risk classifier** achieving 93 percent accuracy on years of historical unstructured financial market data using TF-IDF and BERT feature embedding, and documented the analytical methodology with SOPs co-authored alongside subject matter experts for audit readiness.
- **Led a 4-analyst Agile team** across concurrent analytical workstreams, reviewing outputs, maintaining data governance documentation, and delivering against sprint targets.

Junior Data Analyst | Reporting and Analytics Automation | Decimal Point Analytics Apr 2021 to Jun 2022 | Nashik, India

- Gathered requirements from Product, Finance, and Operations stakeholders, defined KPI frameworks, and translated business needs into SQL queries, Power BI dashboards, and Tableau reports for senior leadership decision-making.
- Performed root cause analysis on metric discrepancies, corrected SQL data models, and deployed monitoring alerts reducing operational reporting errors by 30 percent.
- Developed automated KPI monitoring Power Automate workflows that trigger instant alerts on Teams to Account Executives when high-value client usage drops to a critical threshold, streamlining the churn mitigation process.

Finance Analyst Intern | Decimal Point Analytics | Oct 2020 to Apr 2021 | Nashik, India

- Extracted and structured financial data from client documents. Built Python and Advanced Excel models (VBA, pivot tables, power pivots, array formulas) for financial ratio analysis, trend reporting, and variance analysis with reconciliation checks prior to stakeholder delivery.

KEY PROJECTS

Customer Lifetime Value and Cross-Sell Predictor

Tech Stack: Python, Snowflake, dbt, Apache Airflow, LightGBM, Cox PH Survival Analysis, HDBSCAN, XGBoost, Power BI, DAX, Tableau, Streamlit, Azure OpenAI, DSPy, n8n

End-to-end commercial analytics platform for corporate travel: CLV forecasting, churn timing, customer segmentation, cross-sell propensity, Power BI executive dashboard, and AI-assisted insight automation.

- Scoped the analytical problem into four distinct business questions, defined data requirements and success metrics for each before building, and delivered a Snowflake Modern Data Stack with dbt transformation and Apache Airflow orchestration as the reliable data foundation.
- Built LightGBM CLV forecast (R squared 0.61), Cox PH churn timing model (C-Index 0.89), XGBoost cross-sell propensity (AUC 0.92 plus), and HDBSCAN customer segmentation (Silhouette 0.77). A/B experimentation validated 10 to 20 percent engagement lift (p below 0.0001), quantifying \$800K annual revenue opportunity.
- Built 6-page Power BI strategic dashboard with 30 plus DAX measures and time-intelligence for leadership reporting. Applied AI-assisted narrative generation via Azure OpenAI constrained by deterministic ML outputs, with human validation of every recommendation before stakeholder delivery.

Expense Analytics and Policy Compliance

Tech Stack: Python, SQL, PostgreSQL (Star Schema), Power BI, DAX, Tableau, LangGraph, Vertex AI (Gemini 1.5 Pro), Isolation Forest, DBSCAN, Great Expectations, Apache Airflow, Docker, n8n

Corporate expense compliance analytics platform: 21-signal fraud detection, ML anomaly detection, 6-page Power BI executive dashboard, and AI-enabled self-serve data investigation. \$6.25M projected impact.

- Translated a corporate compliance challenge into a structured analytical framework with 21 behavioural and policy signals across transaction-level, employee-level, and approver-level dimensions — the business requirements definition phase that determined what to measure and why before any modelling began.
- Applied Isolation Forest and DBSCAN anomaly detection alongside chi-squared testing and odds ratio analysis to quantify \$6.25 million projected annual savings across 5 operational levers, with findings presented in a structured executive narrative for CFO and operations leadership.
- Built 6-page Power BI executive dashboard with DirectQuery and custom DAX DateTable, and deployed a LangGraph AI investigation agent enabling audit and compliance teams to query data in natural language — reducing analytical turnaround from hours to minutes while maintaining responsible AI guardrails and human review of all outputs.

Player Churn Prediction and Engagement Optimisation Platform

Tech Stack: Python, PySpark, XGBoost, SQL, A/B Testing, Statistical Power Analysis, FastAPI, Docker

Customer retention analytics platform: predictive churn identification, ROI-gated intervention targeting, A/B experimentation, and measurable engagement impact.

- Redefined churn as a predictive leading indicator rather than a retrospective label, aligning the analytical framework with the actual business decision the team needed: who to contact before they leave, not who already left.
- Designed ROI-gated A/B interventions targeting high-LTV at-risk customers. Statistical power analysis validated 10 to 20 percent relative engagement lift at p below 0.0001, with \$800K annual revenue opportunity quantified and communicated as a structured business case to stakeholders.

FMCG Retail Lakehouse ETL Data Engineering Project

Tech Stack: Databricks, PySpark, Python, SQL, Amazon S3, Delta Lake, Medallion Architecture, dbt, Great Expectations, Unity Catalog, Databricks Genie, Power BI, Tableau

Post-merger FMCG retail data consolidation: Medallion Architecture on Databricks, PySpark distributed processing, and Databricks Genie self-serve analytics for business teams.

- Consolidated two FMCG companies post-merger into a unified Medallion Architecture lakehouse on Databricks: Bronze for raw S3 ingestion, Silver for PySpark schema harmonisation and quality validation, Gold for dimensional KPI models. Databricks Genie enabled natural language self-serve analytics eliminating analyst dependency for routine business questions.

EDUCATION

Data Science and Machine Learning Program | Scaler (powered by InterviewBit)

Feb 2023 to Jun 2025

28-month module-gated program with written examinations and technical interviews. Modules: Statistics, Applied Data Science, Supervised Learning, Unsupervised Learning, NLP, Time Series, Computer Vision, Recommender Systems, Deep Learning, MLOps. Instructors from Google, Amazon, LinkedIn, and MasterCard.

Bachelor of Commerce | Accounting and Finance | Heramba Chandra College, Kolkata

2015 to 2018

CERTIFICATIONS AND CREDENTIALS

Professional Certificates:

IBM Data Science Professional Certificate (Coursera and IBM), IBM Machine Learning Professional Certificate (Coursera and IBM), Probability and Statistics for Machine Learning and Data Science (DeepLearning.AI and Stanford Online)

Scaler Module Certificates:

Advanced Supervised Learning, Unsupervised Learning, Natural Language Processing, Computer Vision, Time Series Analysis, Recommender Systems

Specialised Credentials:

Model Context Protocol Advanced Topics (Anthropic), Building Ambient Agents with LangGraph (LangChain), Hugging Face Agents Course Certificate of Excellence, Generative AI Essentials for Data Science (IBM), Explainable AI in Python (DataCamp), Developing AI Systems with the OpenAI API, SQL Advanced (HackerRank)